CDFD Runtime Paper 09: Tri-Regime Bioenergetics and Life-Number Diagnostics

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Abstract

Executive synthesis. This paper connects CDFD Runtime biology to the tri-regime Life Number Λ. The model tracks energy input, electron transport, proton coupling, stabilization load, relaxation time, and maintenance cost. The local OOL phase experiment supplies a candidate simulation boundary near $\Lambda = 1$. The correct claim is not that the runtime has solved the origin of life. The correct claim is that it provides a reproducible way to test life-like throughput thresholds under explicit model assumptions.

Greek-symbol convention. Unless a manuscript states a tighter equation-local meaning, Greek symbols are local CDFD variables or coefficients, not universal constants. Here Φ denotes driving flux, Ψ_s denotes the adaptive operating ratio, Ω denotes a domain or state space, and Λ denotes the Life Number where used. The coefficients α, β, and γ denote accumulation or reinforcement, relaxation or decay, and diffusion, coupling, or redistribution. The symbols δ and ϵ denote perturbation or tolerance scales, while η, λ, μ, ν, ρ, σ, τ, and ϕ denote local efficiency, rate, baseline, frequency, density or correlation, noise or variance, time-scale, and phase or field terms. Subscripts restrict any coefficient to the named pathway, tissue, domain, or experiment.

Runtime evidence path

Prior CDFD mathematics $\longrightarrow$ CDFL/runtime state $\longrightarrow$ bounded output $\longrightarrow$ falsification target.

Figure 1: Conceptual schematic for the runtime papers. The engine translates the mathematical frame from the prior CDFD papers into executable CDFL/runtime diagnostics; candidate outputs remain tied to explicit tests before any stronger claim is made.

1 Prior CDFD Basis

This manuscript does not rederive the mathematical core of CDFD. It uses the Mujjabi Adaptive Operating Ratio and the constrained-vacuum notation introduced in Part I [1], the torus-knot hierarchy and boundary-mode work from the Part I sequence [2, 3], the public balance law developed in the Part I capstone [4], the Part I transport tests [5], and the Life Number and tri-regime biology bridge from Part II [6]. The runtime papers therefore act as an execution layer: they keep the mathematics connected to commands, outputs, falsification tests, and domain-specific limits.

2 Biology as Bounded Throughput

In this runtime, life-like behavior is modeled as persistent organized flow. The basic local operating ratio is still

$$\Psi_s = \frac{\Phi}{C} S M_s. \tag{1}$$

For origins-of-life questions, however, a system-wide viability number is also necessary because local Ψ_s can be high in one region and low elsewhere.

3 The Life Number

The Life Number is

$$\Lambda = \frac{\Phi \sigma_e \sigma_p \tau_{\mathrm{relax}}}{S_{\mathrm{stab}} E_{\mathrm{maintenance}}}, \tag{2}$$

where Φ is energy input, σ_e is electron-transport effectiveness, σ_p is proton-coupling or coherence effectiveness, τ_{relax} is relaxation time, S_{stab} is stabilization or dissipation burden, and $E_{\mathrm{maintenance}}$ is maintenance energy.

The working labels are:

- $\Lambda < 1$: decay-dominated;
- $\Lambda \approx 1$: near-critical proto-biological window;
- $\Lambda > 1$: sustained throughput in the model.

4 Runtime Implementation

The implementation lives in `engine/origins_of_life.py` and is called from `engine/biology.py`. The runtime stores channel summaries in `state.meta` keys such as `ool_C_input`, `ool_sigma_e`, `ool_sigma_p`, `ool_S_stability`, `life_number`, and `throughput_J`.

5 Experiment: OOL Phase Boundary

The OOL phase script sweeps a 50 by 50 parameter grid over energy input and geochemical constraint. The public report describes candidate points near $\Lambda = 1$, and the selected HDF5 output stores the grid.

The useful result is a model diagnostic: the runtime can locate and save a phase-boundary surface in the declared toy parameter space. The result still needs testing against laboratory chemistry, geochemical constraints, and independent origin-of-life models before any empirical claim is made.

6 Biology Discovery Scripts

The broader biology experiments include:

- `experiments/run_biology_discovery.py`: metabolic saturation, chronic drift, and tri-regime throughput;
- `experiments/run_bio_mujjabi_tests.py`: parasitic-drain and cycling-retention tests;
- `experiments/run_drug_synergy_predictions.py`: same-axis versus cross-axis perturbation toy trajectories.

The public `runtime/diagnostics.py` module exposes guarded aromatic source-mix rows (Part II Paper 7), photochemical endpoint status (Paper 11), and the Life Number supply guardrail. The primary export path is `python cdfd.py diagnostics`; the optional Streamlit studio and `cdfd.py demo origins_of_life --source-scenario <name>` surface the same values.

These are model diagnostics and hypothesis generators. They are not care instructions or validated use rules.

7 Falsification Conditions

The tri-regime model fails or requires revision if:

- measured prebiotic systems show no reproducible threshold near the proposed Λ proxy;
- σ_e or σ_p cannot be measured or approximated in a domain-specific way;
- the boundary is indistinguishable from a simpler energy-only model;
- outputs become non-finite under reasonable parameter stress.

8 Conclusion

Tri-regime bioenergetics gives CDFD a bridge from local operating ratios to system-level viability. Its strength is not rhetorical certainty; its strength is that it produces explicit variables, runnable scripts, saved outputs, and failure conditions.

References

[1] Steve Bico Mujjabi, MD. *Vortex Stability in a Constrained Transport Vacuum and the Origin of the Fine-Structure Constant.* CDFD Part I Paper I, preprint, 2026.

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[6] Steve Bico Mujjabi, MD. *CDFD Part II: Origins of Life and Tri-Regime Bioenergetics.* Public release archive, 2026, `https://doi.org/10.5281/zenodo.20264779`.